

ARTIFICIAL NEURAL NETWORKS APPLICATION IN TIME SERIES WATER DISCHARGE FOR THE PREDICTION OF FLOODING IN MOHAWK RIVER, NEW YORK

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Abstract

A time series model enhanced with an artificial neural network model has been used for the prediction of flood events in Mohawk River, New York. For the prediction of the water discharge time series, climatic and hydrogeological variables have been used in the analysis. To separate the different signals in the time series a low pass filter has been applied in the raw data. The Kolmogorov-Zurbenko (KZ) filter has been used for the decomposition of the time series into the long-, seasonal-, and short-term component. Each time series component has been modeled using the multiple linear regression model (MLR), and the artificial neural network (ANN) model. The time series decomposition combined with the ANN model provide the flood prediction of the water discharge time series and lead to a forecasting up to 96% when the climatic and hydrogeological variables have been used in the model. Although the MLR model provides a lower performance, it retains the advantage of the physical interpretation of the water discharge time series, while the ANN model shows as an outstanding prediction model.

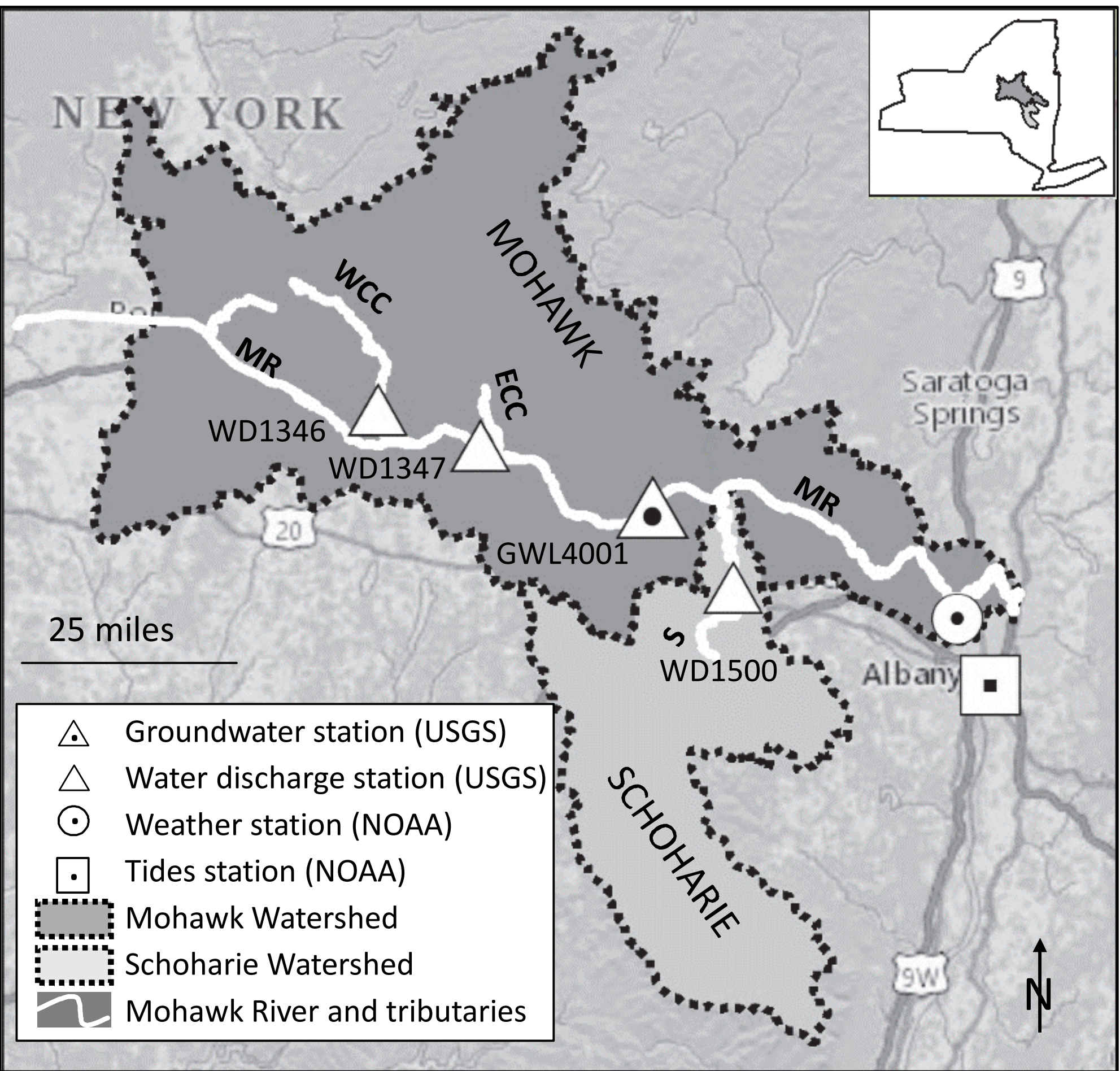


Figure 1: Mohawk watershed map with locations of the monitoring stations and associated variables used in the MLR and ANN.

Introduction

Flood events are natural hazards yielding severe impact to socio-economic structure [1], and in Upstate New York along the Mohawk River and other tributaries, prediction efforts of those flood events have been made [2]. Flood forecasting research is required in highly occupied regions such as in New York State. We aim to separate individual patterns and decrease interferences originating from covariance structures between the components of the time series. We compare linear (MLR) with non linear (ANN) modeling toward improving a flood warning system for several sections of Mohawk Watershed in New York (Figure 1).

Methods

- Data mining and cleaning data sets of climatic and hydrological variables such as daily water discharge, wind speed, temperature, tides, precipitation, and groundwater level.
- Decomposition of time series and separation of the different signals with minimum interferences between the components of the time series [3,4]
- Raw data were decomposed to long-term trend (fluctuations longer than a given threshold), seasonal-term (intraannual fluctuations), and a short-term component (short-term variations) [5].
- Multiple Linear Regression and Artificial Neural Network (Figure 2) development was performed.

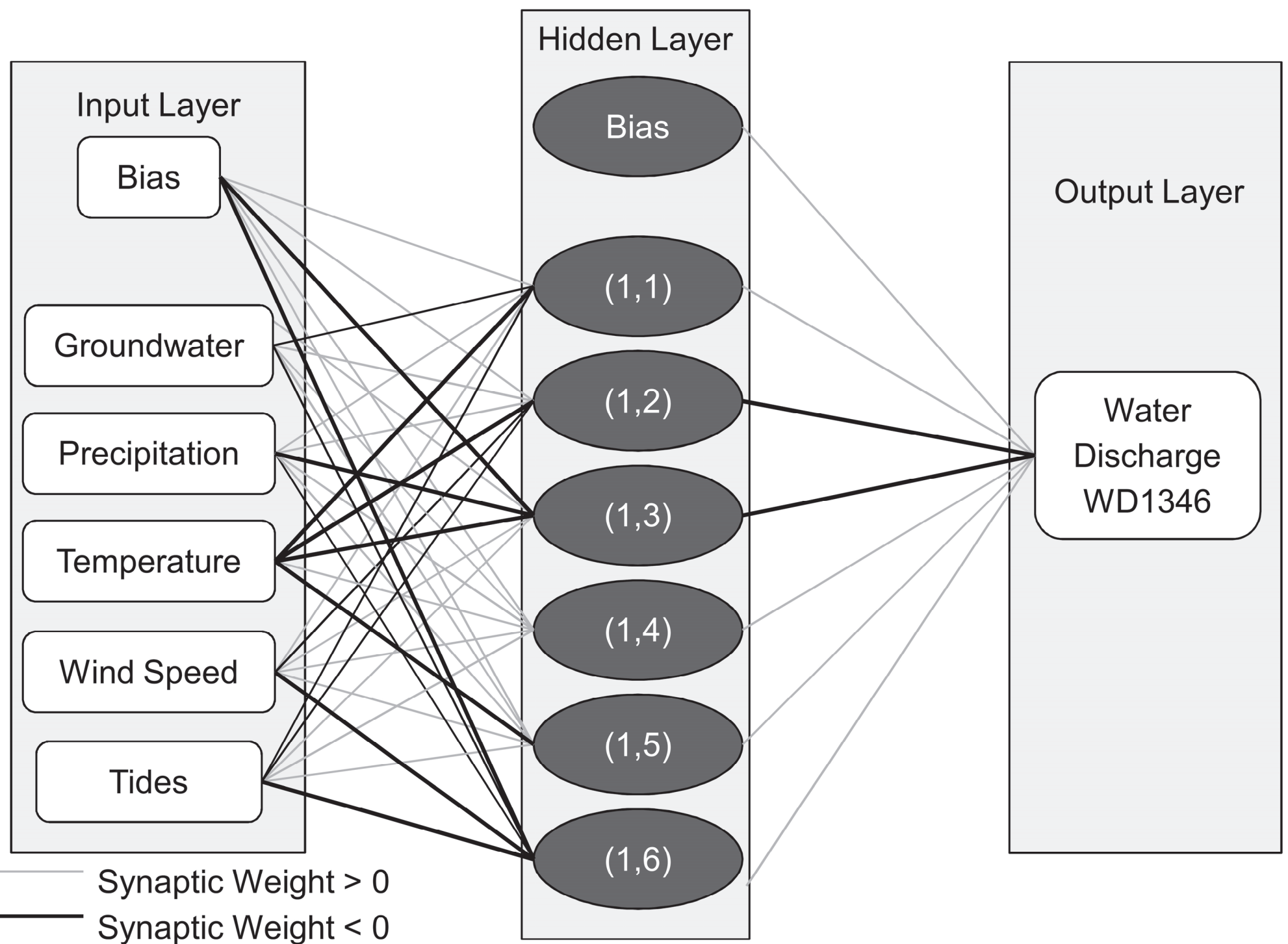


Figure 2: The feedforward artificial neural network of the raw data from the USGS station WD1346 using a hidden layer transfer function of hyperbolic tangent and an output layer for the prediction of water discharge [6].

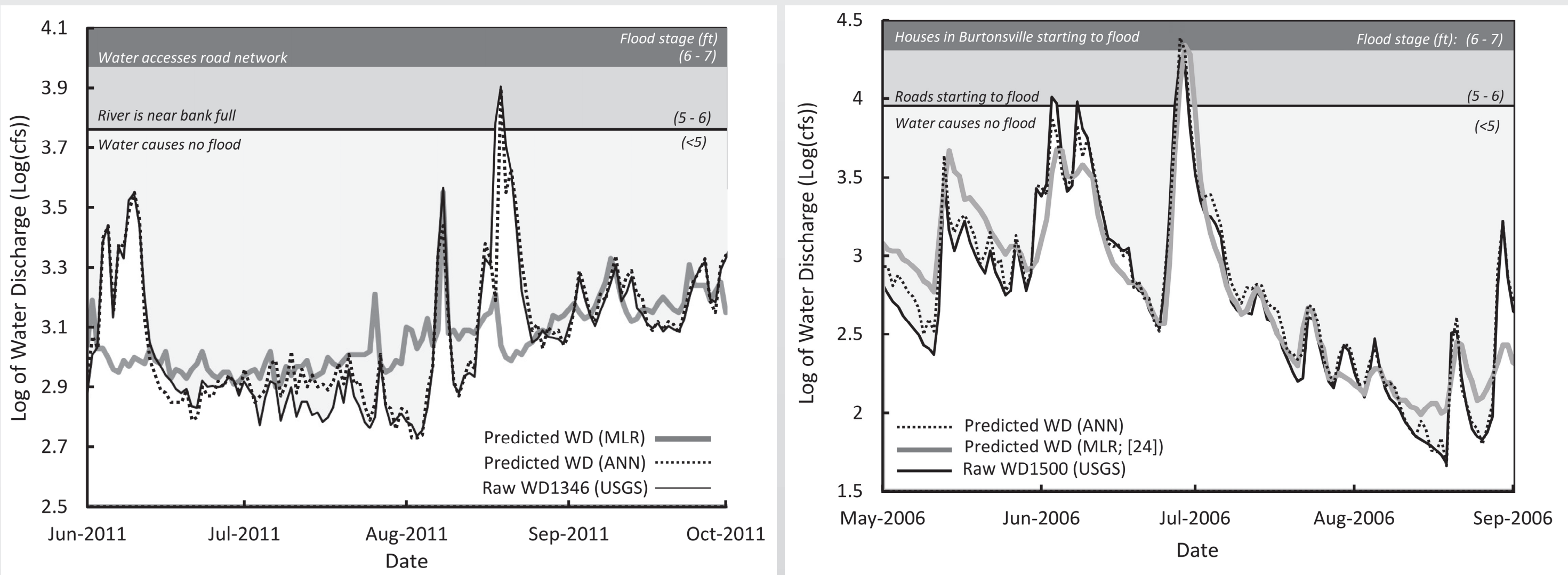


Figure 3: (left picture) The predicted MLR and ANN values plotted against the raw values of the water discharge time series for the station WD1346 (left) and WD1500 (right). Significant peaks reflect the Irene storm (21 August to 28 August 2011; left) in West Canada Creek at Herkimer, NY and the Mid-Atlantic United States flood of 2006 (25 June to 5 July 2006) in Schoharie Creek, NY.

Results

Table 1: Results and performance described by coefficient determination and total explanation from the application of MLR and ANN in flood prediction

Station	Raw	Long	Seasonal	Short	Station	Raw	Long	Seasonal	Short
Multiple Linear Regression R^2					Artificial Neural Network R^2				
WD1346	48.7	73.5	93.1	70.6	WD1346	53.4	97.6	93.1	94.9
WD1347	47.6	73.6	92.1	68.3	WD1347	55.4	97.6	95.1	95.3
WD1500	59.0	83.3	91.2	34.1	WD1500	60.7	98.0	94.5	94.1

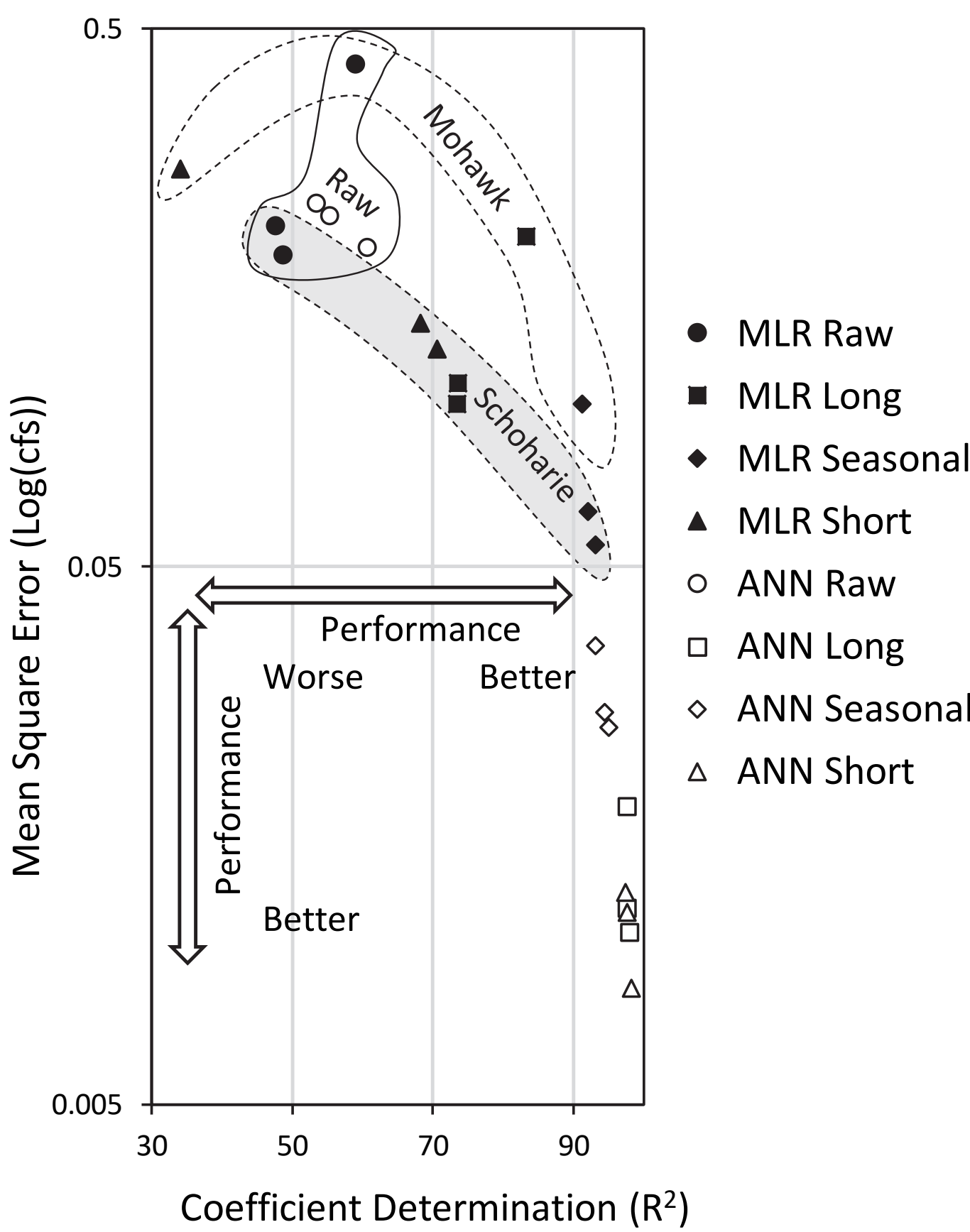


Figure 4: The MSE as a function of the coefficient determination (R^2) for the raw and the decomposed time series data using the MLR and the training ANN model.

Discussion

- ANN shows a much better prediction performance than MLR (Table 1; Figure 3)
- MLR allows better approach to the physical interpretation of the hydrogeological model (Figure 4)

Conclusion

The hybrid model application of the time series decomposition and ANNs has been shown to be an efficient model of flood hazard forecasting, and it presents an improved prediction model for a flood hazard up to 96%. However, the MLR allows physical interpretation of the prediction model.

References

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